

AI-BASED MOVIE RECOMMENDATION SYSTEM USING JAVA

1. Abstract

Recommendation systems play a vital role in modern applications by suggesting relevant items to users based on their interests and preferences. This project implements a simple AI-Based Movie Recommendation System using Java. The system accepts the user's preferred movie genre and recommends a list of movies belonging to that genre. The project utilizes Java Collections Framework, specifically HashMap and ArrayList, to store and manage movie data. The recommendation logic is based on matching user preferences with available movie categories.

This project demonstrates the basic principles of recommendation systems and intelligent decision-making used in real-world applications.

2. Objective

The objectives of this project are:

- To understand the concept of recommendation systems.
- To develop a simple AI-based application using Java.
- To learn the use of HashMap and ArrayList.
- To recommend movies based on user preferences.
- To demonstrate intelligent decision-making using predefined datasets.

3. Software Requirements

Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB or above
- Storage: 500 MB free space

Software Requirements

- Operating System: Windows/Linux/MacOS
- Java Development Kit (JDK 17)
- Command Prompt (CMD)
- Notepad or any Java IDE

4. Introduction

Artificial Intelligence is widely used in recommendation systems to suggest products, movies, music, and other content based on user preferences. Popular platforms such as Netflix, Amazon, YouTube, and Spotify use recommendation engines to improve user experience.

In this project, a simple recommendation system is developed using Java. The user selects a preferred movie genre, and the system recommends movies belonging to that category. Although this project uses a predefined dataset, it demonstrates the fundamental concept behind recommendation systems.

5. Concepts Used

Artificial Intelligence

Artificial Intelligence enables systems to make decisions and recommendations based on available information.

Recommendation System

A recommendation system suggests relevant items to users based on preferences or behavior.

HashMap

HashMap stores movie genres as keys and movie lists as values.

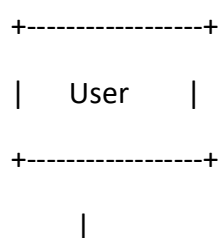
ArrayList

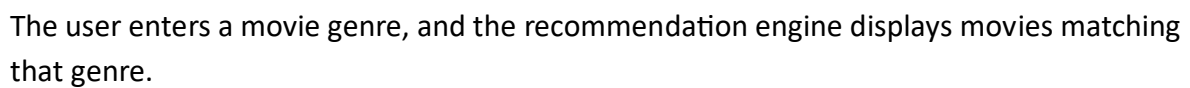
ArrayList stores multiple movie names under each genre.

User Input

The Scanner class is used to accept user preferences.

6. System Architecture





1. Create a HashMap to store movie genres and movie lists.
2. Add predefined movie data to the HashMap.
3. Display available genres.
4. Accept user input for preferred genre.
5. Check whether the genre exists.
6. If found, display recommended movies.
7. Otherwise, display an error message.

8. End the program.

8. Source Code

```
import java.util.*;

public class MovieRecommendationSystem {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        HashMap<String, ArrayList<String>> movies = new HashMap<>();

        movies.put("Action", new ArrayList<>(Arrays.asList(
            "Avengers",
            "John Wick",
            "Mad Max",
            "Mission Impossible"
        )));

        movies.put("Comedy", new ArrayList<>(Arrays.asList(
            "3 Idiots",
            "Jathi Ratnalu",
            "Dhamaal",
            "Golmaal"
        )));

        movies.put("Drama", new ArrayList<>(Arrays.asList(
```

```
        "The Pursuit of Happyness",  
        "Forrest Gump",  
        "Jersey",  
        "Maharshi"  
    ));
```

```
movies.put("SciFi", new ArrayList<>(Arrays.asList(  
    "Interstellar",  
    "Inception",  
    "The Matrix",  
    "Avatar"  
)));
```

```
System.out.println("Available Genres:");  
System.out.println("Action");  
System.out.println("Comedy");  
System.out.println("Drama");  
System.out.println("SciFi");
```

```
System.out.print("\nEnter your preferred genre: ");  
String genre = sc.nextLine();
```

```
if (movies.containsKey(genre)) {
```

```
    System.out.println("\nRecommended Movies:");
```

```
    ArrayList<String> recommendations = movies.get(genre);
```

```
        for (int i = 0; i < recommendations.size(); i++) {  
            System.out.println((i + 1) + ". " + recommendations.get(i));  
        }  
  
    } else {  
  
        System.out.println("\nSorry! Genre not found.");  
    }  
  
    sc.close();  
}  
}
```

9. Sample Output

Available Genres:

Action

Comedy

Drama

SciFi

Enter your preferred genre: Action

Recommended Movies:

1. Avengers
2. John Wick
3. Mad Max
4. Mission Impossible

(Attach your CMD output screenshot here.)

10. Applications

- Netflix Movie Recommendations
- Amazon Product Recommendations

- YouTube Video Suggestions
- Spotify Music Recommendations
- Online Learning Platforms
- E-commerce Websites

11. Advantages

- Easy to implement.
- Improves user experience.
- Provides personalized suggestions.
- Efficient data retrieval using HashMap.
- Demonstrates AI concepts in a simple manner.

12. Limitations

- Uses a predefined dataset.
- Does not learn from user behavior.
- Recommendations are based only on genre selection.
- Does not use machine learning algorithms.

13. Future Enhancements

- Integrate Machine Learning algorithms.
- Use user ratings and feedback.
- Store data in a database.
- Add graphical user interface (GUI).
- Support personalized recommendations.
- Connect to online movie databases through APIs.

14. Conclusion

The AI-Based Movie Recommendation System successfully demonstrates the basic working of a recommendation engine using Java. The system accepts user preferences and provides relevant movie suggestions. The project highlights the importance of recommendation systems in modern applications and serves as a foundation for more advanced AI-powered systems.

15. Viva Questions and Answers

Q1. What is a recommendation system?

A recommendation system is a software application that suggests relevant items to users based on preferences, interests, or behavior.

Q2. Why is this project called AI-Based?

It mimics intelligent decision-making by recommending movies according to user preferences.

Q3. Which data structures are used in this project?

HashMap and ArrayList.

Q4. What is the purpose of HashMap?

HashMap stores movie genres as keys and corresponding movie lists as values.

Q5. What is the purpose of ArrayList?

ArrayList stores multiple movie names under a genre.

Q6. What is the role of Scanner?

Scanner is used to accept user input from the keyboard.

Q7. What are real-world examples of recommendation systems?

Netflix, Amazon, YouTube, Spotify, and Flipkart.

Q8. Can this project be improved?

Yes, by integrating machine learning algorithms and user behavior analysis.

Q9. What are the benefits of recommendation systems?

They improve user experience, increase engagement, and provide personalized suggestions.

Q10. What is the future scope of recommendation systems?

Recommendation systems can be enhanced using Artificial Intelligence, Machine Learning, Big Data, and Cloud Computing technologies.

